Technical Design Document (TDD)

Intelligent Customer Support Agent for E-Grocery using Streamlit & Snowflake Cortex AI

DOCUMENT VERSION	TARGET STACK	EMBEDDING MODEL	SECURITY TIER
1.0.0 (Production)	SiS, Cortex AI, Snowpark	e5-base-v2 (768-dim)	Snowflake RBAC / In-DB

1. EXECUTIVE PROBLEM ALIGNMENT & SCOPE

High daily order volumes of perishable goods result in high tier-1 customer service costs and wait times. This Technical Design Document specifies an autonomous, data-aware AI Agent built natively inside Snowflake using **Streamlit in Snowflake (SiS)** and **Snowflake Cortex AI**. The system unifies transactional lookup, unstructured policy retrieval (RAG), and sentiment-driven feedback logging into a single zero-data-leakage architecture.

2. SYSTEM OBJECTIVES & SUCCESS METRICS MAPPING

Core Functional Objectives

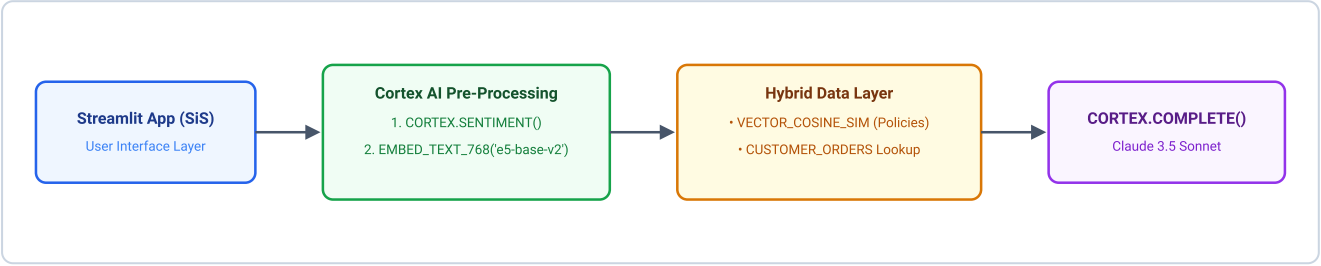
- **Product & Policy RAG:** Natural language retrieval for dietary info, stock, and return rules.
- **Return/Refund Automation:** Validation of order status and perishable return window (48 hrs).
- **Sentiment Logging:** Real-time score recording in `CUSTOMER_FEEDBACK`.

Target Target Success Metrics

- **Deflection Rate:** Reduce routine ticket volume by **≥ 40%**.
- **Resolution Latency:** Maintain sub-5-second query processing (Target: **< 3.0s**).
- **Data Grounding: 0% hallucination** on orders or amounts via direct SQL & vector context.

3. END-TO-END SYSTEM ARCHITECTURE DIAGRAM

All data flow, embedding calculation, and LLM reasoning are executed within Snowflake security boundaries:

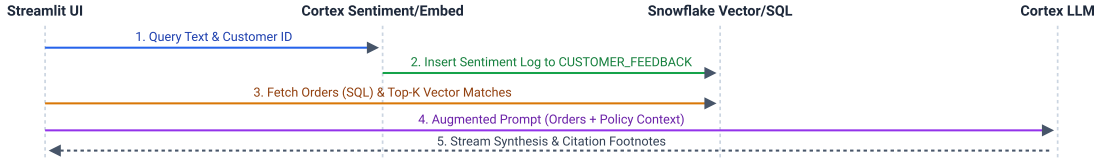


4. DETAILED DATABASE & VECTOR SCHEMA SPECIFICATIONS

Table Name	Type / Indexing	Primary Key / Columns	Data Purpose & Governance
`PRODUCT_CATALOG`	Relational Table	`PRODUCT_ID`, `NAME`, `DIETARY_INFO`, `STOCK_QTY`	Stores product attributes, stock counts, and allergen information.
`POLICY_DOCUMENT_CHUNKS`	Vector Table	`DOC_NAME`, `CHUNK_TEXT`, `CHUNK_EMBEDDING VECTOR(FLOAT,768)`	Contains vectorized policy chunks generated via Cortex `EMBED_TEXT_768`.
`CUSTOMER_ORDERS`	Transactional DB	`ORDER_ID`, `CUSTOMER_ID`, `ORDER_DATE`, `STATUS`	Filtered by `CUSTOMER_ID` to enforce strict RBAC data isolation.
`CUSTOMER_FEEDBACK`	Audit Analytics Log	`FEEDBACK_ID`, `CUSTOMER_ID`, `SENTIMENT_SCORE`, `CATEGORY`	Logs user input, Cortex sentiment scores (-1.0 to 1.0), and feedback categories.

5. COMPONENT EXECUTION & RAG SEQUENCE DIAGRAM

The sequence below illustrates the lifecycle of a query from reception to grounded response synthesis:



6. CORE CORTEX FUNCTIONS IMPLEMENTATION LOGIC

Vector Similarity SQL Query
Uses `e5-base-v2` to compute cosine distance across policy chunks:

```
SELECT DOC_NAME, CHUNK_TEXT,
VECTOR_COSINE_SIMILARITY(CHUNK_EMBEDDING,
q_emb) AS SCORE
FROM POLICY_DOCUMENT_CHUNKS, query_vec
ORDER BY SCORE DESC LIMIT 3;
```

Sentiment & Category Pipeline
Evaluates feedback sentiment (-1 to 1) and categorizes queries:

```
SELECT SNOWFLAKE.CORTEX.SENTIMENT(:msg);
SELECT SNOWFLAKE.CORTEX.COMPLETE('llama3.1-
8b',
'Categorize into [Delivery, Return, Stock]:
' || :msg);
```

7. SECURITY, ROLE-BASED ACCESS CONTROL (RBAC) & GUARDRAILS

Data Isolation & RBAC Rules

- **Session Scoping:** Every transactional SQL query filters strictly by active `CUSTOMER_ID` context.
- **Boundary Protection:** Zero data egress; embedding vectors and LLM context stay within the Snowflake warehouse.
- **Sanitization:** Input text is escaped against standard SQL injection vectors.

Automated Escalation Guardrails

- **Frustration Threshold:** Sentiment scores `< -0.50` initiate immediate routing to human support queues.
- **RAG Relevance Floor:** Vector cosine similarity `< 0.35` triggers an explicit "Insufficient Context" response.
- **Refund Cap:** Automated returns capped at $50 per transaction without manual review.

8. VERIFICATION & METRIC VALIDATION MATRIX

Objective Domain	Technical Validation Method	Target KPI Benchmark
Policy Query Accuracy	Grounding via top-3 vector chunks with document citations.	≥ 95% Precision
Return Automation	SQL validation of 48-hr perishable delivery timestamp window.	100% Policy Compliance
Sentiment Capture	Synchronous write to `CUSTOMER_FEEDBACK` on each input.	100% Audit Logging
End-to-End Latency	Optimized warehouse size (XSMALL) and single-pass prompts.	< 3.0 Seconds Total

9. DEPLOYMENT & MAINTENANCE OPERATIONAL PLAN

The solution is deployed via standard Snowflake SQL Worksheets and Streamlit in Snowflake environments. Daily policy updates are automated through a scheduled Snowflake Task that parses new documents added to `@ONLINE_POLICY_STAGE` and updates `POLICY_DOCUMENT_CHUNKS` using Cortex AI vector generation.